Programming Fundamentals

Week 8 — Importing Python Modules

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Why Do We Use `import`?

Imagine writing every tool you need from scratch: random numbers, JSON reading, maths, and every function from previous projects.

`import` lets one Python file use code from another module.

A **module** is usually a Python file containing functions, variables or classes.

We have we already imported in our programs - (`json`, `random`, `pathlib`)

Three Places Python Code Can Come From

Source	Examples	How we get it
Standard library	`random`, `json`, `math`	Included with Python
Third-party packages	`requests`, `numpy`	Installed separately, often with `pip`
Our own modules	`calculator.py`	We create the `.py` file

Important: `import` does not install a missing package.

Example 1 — Import a Standard Library Module

```
import random

number = random.randint(1, 6)
print("Dice roll:", number)
```

- `random` is the module.
- `randint()` is a function inside that module.
- The dot means **look inside this module**.

Try: Change `6` to `20`. What values could be printed?

Two Ways to Import

```
import math
print(math.sqrt(25))

from math import sqrt
print(sqrt(25))
```

Both print `5.0`.

- `import math` keeps the module name visible.
- `from math import sqrt` brings that name directly into this file.

Beginner tip: `import math` often makes the source of a function clearer.

Example 2 — Create Our Own Module

File 1: `calculator.py`

```
def add(a, b):
    return a + b

def divide(a, b):
    return a / b
```

This file defines reusable functions. It does not need to ask for input or print a menu.

Example 2 — Use It from Another File

File 2: `main.py`

```
import calculator

answer = calculator.add(10, 5)
print("Total:", answer)
```

Folder:

```
MyProject/
    main.py
    calculator.py
```

Both files are in the same folder. Python can normally find `calculator.py` when running `main.py`.

Import Just One Function

```
from calculator import add

answer = add(10, 5)
print("Total:", answer)
```

Compare:

- `import calculator` ⟶ `calculator.add(10, 5)`
- `from calculator import add` ⟶ `add(10, 5)`

Question: Which version makes it clearer where `add` was defined?

What Runs When We Import?

Python executes a module's top-level statements when it is first imported in a process.

In `greetings.py`:

```
print("Module is loading")

def say_hello():
    print("Hello!")
```

Importing `greetings` prints `Module is loading` — even without calling `say_hello()`.

Good practice: Keep reusable functions in modules; put interactive program startup in `main.py`.

Optional — The `__name__` Guard

```
def add(a, b):
    return a + b

if __name__ == "__main__":
    print(add(2, 3))
```

- Running this file directly: the test prints `5`.
- Importing this file: the test does **not** run.

You do not need to memorise it today.